



GCE A LEVEL

1420U50-1D



S26-1420U50-1D

To be opened on WEDNESDAY, 18 MARCH 2026 by TEACHERS

PHYSICS – A2 unit 5
Practical Examination

Experimental Task
TESTS 1 and 2

SETTING UP INSTRUCTIONS

CONFIDENTIAL

This document should be stored securely by the exams officer when not in use by the teacher/technician.

This document must not be stored within the science department when not in use by the teacher/technician.

Its contents should not be divulged except to those concerned with the preparation of the assessment.

1420U501D
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Answer all questions.

- (a) Your task is to carry out an investigation to see how the resistance, R of a filament bulb varies with the potential difference, V across it.

You are provided with the following equipment

12 V d.c. power supply
12 V, 24 W filament bulb
Voltmeter
Ammeter
Variable resistor/potentiometer
4mm leads and crocodile clips
Push to make switch

The relationship between resistance, R and potential difference, V can be expressed by:

$$R = kV^n,$$

where k and n are unknown constants.

Write out a plan of how you will obtain sufficient readings to investigate this relationship.

Include a circuit diagram with your plan.

Repeat readings are not needed for this task.

Space for risk assessment on next page.

Risk assessment

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The remaining parts of the examination paper are not required for the purpose of checking the setting up of the task.

TEST 2

All of the above applies except power of a filament bulb varies with the potential difference across it.

$$P = kV^n$$

AWARDING OF TEACHER ASSESSED MARKS

Please ensure that these marks are recorded on the front of each candidate's paper.(a)

(i) **The mark is to be awarded if the candidate has set up the circuit correctly**

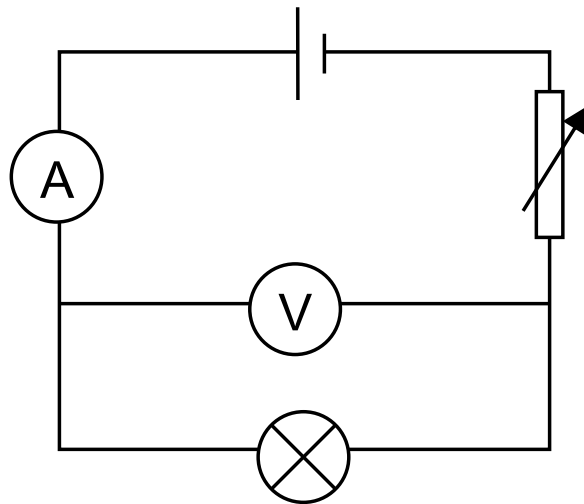
If the candidate incorrectly sets up the circuit, **do not award the mark**, but give the candidate the information sheet (on the next page) to assist them in setting up the experiment. **However**, candidates are allowed to request that the teacher take down the circuit, and in this case, do not give the candidate the instruction sheet.

(ii) **This mark is to be awarded if the candidate states that there is a potential risk of hot lamp.**

(Hazard)	(Risk)	(Control measure)
Hot lamp (can burn)	Touching the lamp and burning skin	Leave the lamp to cool before moving it / avoid touching

INFORMATION SHEET

Sufficient copies of this information sheet should be made in advance.
The information sheet is the same for Tests 1 and 2.



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